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Nayan Bhatia

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Education

University of California, Santa Cruz Sep 2021 – Jun 2027 (exp.)
PhD Candidate in Computer Science & Engineering (transitioned from M.S)
Inter-Networking Research Group (i-NRG Lab) under Prof. Katia Obraczka

K.J. Somaiya Institute of Engineering and IT, University of Mumbai Aug 2017 – Jun 2021
B.E. in Computer Engineering

Publications & Preprints

- **N. S. Bhatia**, P. Kocheta, R. Elliott, H. S. Kuttivelli, and K. Obraczka. *Indoor Localization Using Compact, Telemetry-Agnostic, Transfer-Learning Enabled Decoder-Only Transformer*. Accepted as Work-in-Progress (WiP), IEEE PerCom 2026 (CORE A* ranked conference). Preprint: arXiv:2510.11926. Patent Disclosure: UC Case 2026-577-0.
- P. Kocheta, **N. S. Bhatia**, and K. Obraczka. *Pulse-Fi: A Low-Cost System for Accurate Heart Rate Monitoring Using Wi-Fi Channel State Information*. In **IEEE DCOSS-IoT 2025**, Short Paper. PDF.
- H. S. Kuttivelli, L. Krishnaswamy, **N. S. Bhatia**, and K. Obraczka. **IEEE ICDCS 2025 Tutorial**: *Network Simulation Bridge*. Tutorial Link. Supported by NSF POSE Award #2449377.
- H. S. Kuttivelli, S. Sreenivasamurthy, L. Krishnaswamy, **N. S. Bhatia**, and K. Obraczka. *Network Simulation Bridge: Bridging Applications to Network Simulators*. **ACM Q2SWinet**, 2023. PDF.

Research & Projects

Locaris: LLM-based Indoor Localization from Wi-Fi Telemetry 2025 – Present

- Developed **Locaris**, a decoder-only transformer that treats Wi-Fi telemetry (CSI, RSSI, FTM) as a "language" and directly predicts device position.
- Reduced reliance on handcrafted feature engineering and improved cross-environment generalization.
- Achieved sub-meter accuracy on RSSI/FTM and centimeter-level precision on CSI across controlled testbeds.

PulseFi: Contactless Vital-Signs via Commodity Wi-Fi 2024 – Present

- Built a low-cost sensing pipeline on ESP32 and Raspberry Pi devices to extract Wi-Fi Channel State Information (CSI) and estimate multiple vital signs.
- Demonstrated accurate monitoring of **heart rate, breathing rate, and fall detection** using ultra-low-cost commodity hardware.
- Co-led dataset curation and modeling across internal and public benchmarks; validated system against medical-grade sensors.
- Featured in notable outlets including **University of California News**, **Tom's Hardware**, **Yahoo Tech**, and **CNET**, underscoring both academic credibility and global tech media recognition. Media coverage summary: Pulse-Fi Press Coverage.

Experience

University of California, Santa Cruz Oct 2021 – Present
LLM & Platform Engineer

- Developed the **UC Basic Needs (UCBN) platform** (ucbasicneeds.ucsc.edu) and **UC Essential Needs Navigator**, a UC-wide system serving ~300,000 students across 10 campuses; featured by UCSC News.
- Led end-to-end system implementation as primary developer, building the full-stack architecture and AI-powered search experience in collaboration with UC students, staff, and faculty stakeholders.
- Designed the AI-powered Navigator enabling natural language queries with cited results, improving accessibility to critical resources (food, housing, healthcare, etc.) across campuses.

Awards & Honors

- **QB3 Santa Cruz Graduate Innovators Fellowship (2026)**: Awarded for advancing PulseFi, a WiFi-based vital signs monitoring system with commercialization potential in healthcare.
- **Biodiversity Seed & Agroecology Fellowship**: $11,000 award for the UC Essential Needs (UCBN) platform.
- **CITRIS Smart Health Masterclass & Intensive (2025)**: Nominated for a selective 5-day program at UC Berkeley on responsible AI in healthcare and healthtech innovation.
- **Certificate of Recognition (2024, 2025)**: Awarded by UC Basic Needs & Economic Justice Center for exceptional contributions and leadership in student resource initiatives.
- **Technical Program Committee Member**, IEEE LCN 2026.
- **Peer Reviewer**: IEEE Transactions on Mobile Computing (TMC), 2025; IEEE Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS), 2026.
- **Invited Presenter, Edge AI and Vision Alliance**: Presented PulseFi to an international community of edge AI and computer vision practitioners.
- **Invited Presenter, UC Basic Needs Convenings (2022–2026)**: Presented at CHEBNA 2024/2026 (Sacramento), UC GPC Summit (UCLA), and UC Basic Needs Convening (UC Irvine).
- **Young Gladiators, Colosseum Master Class (DARPA/NSF)**: Invited for training on large-scale wireless experimentation using the Colosseum emulator.
- **SIP Mentor (2024–2025)**: Led 4 interns on ML-based indoor localization; ~95%+ test accuracy in lab setting.
- **Teaching Assistant, UCSC**: Courses in Python, C, IoT, Database Systems, and Operating Systems; mentored 100+ students per quarter.
- **Project Judge, CruzHacks 2026**: Evaluated student hackathon projects at UC Santa Cruz's largest annual hackathon.

Technologies

Programming: Python, SQL, C/C++, Java, Scala, Solidity

ML/AI: PyTorch, TensorFlow, Keras, NumPy/Pandas, scikit-learn, Matplotlib

Web: Node.js, React.js, Express, GraphQL, Flask, Django; HTML/CSS/JS

Data/Infra: MySQL, MongoDB, CUDA; Docker, Kubernetes, Nginx, GitHub Actions, CI/CD; AWS/GCP

MLOps/Automation: Jenkins, Ansible, Pytest/PyTorch-Lightning, Selenium, SLURM